MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, SILICON, HIGH POWER, THREE PHASE, FULL WAVE BRIDGE RECTIFIER PART NUMBERS M19500/483-01 THROUGH M19500/483-04

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for three phase bridge rectifiers for use at frequencies of 1 kHz, or less.

1.2 Physical dimensions. See figure 1.

1.3 Maximum ratings.

Part number	V_{RWM}	Bridge 1/ I_{out} @ T_C = 55°C	Bridge 2/ I_{out} @ T_C = 100°C	i_f(surge) t_p = 8.3 ms; I_{out} = 25 Adc
M19500/483-	V (pk)	Adc	Adc	A (pk)
01	200	25	18.5	150
02	400	25	18.5	150
03	600	25	18.5	150
04	800	25	18.5	150

1/ Derate from 25 Adc at 55°C to 18.5 Adc at 100°C (144 mAdc/°C).
2/ Derate from 18.5 Adc at 100°C to 0 Adc at 150°C (370 mAdc/°C).

OPERATING AND STORAGE TEMPERATURE: -65° to +150°C

BAROMETRIC PRESSURE, REDUCED: 8 mmHg

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor devices, General Specification for.

STANDARDS

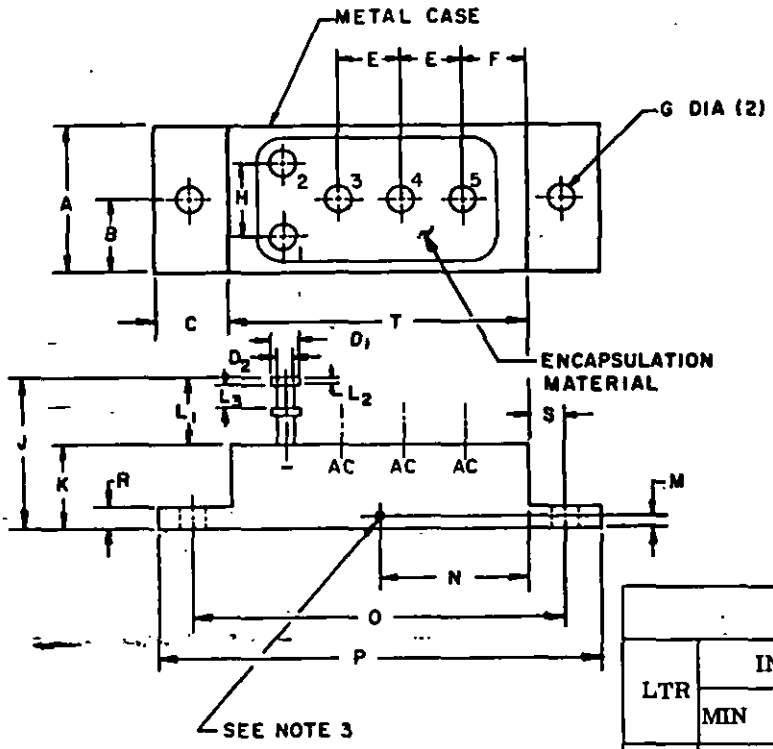
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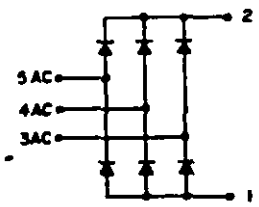
MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.
MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 5961

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LTR	DIMENSIONS				NOTES
	INCH		MILLIMETERS		
	MIN	MAX	MIN	MAX	
A	.730	.770	18.54	19.56	
B	.355	.395	9.02	10.03	
C	.355	.395	9.02	10.03	
D_1	.141	.151	3.58	3.84	
D_2	.108	.118	2.74	3.00	
E	.355	.395	9.02	10.03	
F	.230	.270	5.84	6.86	
G	.149	.189	3.78	4.80	
H	.355	.395	9.02	10.03	
J		.82		20.83	
K	.39	.51	9.91	12.95	
L_1	.240	.320	6.10	8.13	
L_2	.015	.030	.38	.76	
L_3	.100	.125	2.54	3.18	
M	.040	.060	1.02	1.52	
N	.72	.78	18.29	19.81	
O	1.84	1.90	46.74	48.26	
P	2.22	2.28	56.39	57.91	
R	.09	.15	2.29	3.81	
S	.168	.208	4.27	5.28	
T	1.47	1.53	37.34	38.86	

NOTES:
1. Terminals shall be tinned.
2. Polarity shall be marked as shown on drawing.
3. Point at which T_C is read (shall be in metal part of case).

FIGURE 1. Semiconductor device, silicon, high power, three phase, full wave bridge rectifier.

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3. REQUIREMENTS

3.1 General. Requirements shall be in accordance with MIL-S-19500, and as specified herein.

3.2 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-S-19500.

3.3 Design, construction, and physical dimensions. The rectifier bridge shall be of the design, construction, and physical dimensions shown on figure 1.

3.3.1 Internal construction. The rectifier bridge shall consist of a metal and plastic encased assembly of discrete diodes. Each discrete diode shall be in a glass-to-metal, ceramic-to-metal, or fused metal oxide-to-metal hermetically sealed package. The silicon die in each discrete diode shall be metallurgically bonded directly to the terminal pins. The completed assembly of diodes and other internal structures shall be encapsulated in a plastic material which polymerizes to a rigid condition by virtue of a chemical cross-linking mechanism. Discrete diodes for assembly of bridge rectifiers shall be subjected to process and power conditioning (see 4.4.2.1).

3.4 Performance characteristics. Performance characteristics shall be as specified in tables I, II, and III.

3.5 Marking. Marking shall be in accordance with MIL-S-19500, except that the "M" part number (as shown in 1.3) shall be used in lieu of the type designation.

3.5.1 Polarity marking. Polarity shall be indicated as shown on figure 1.

4. QUALITY ASSURANCE PROVISIONS

4.1 General. Quality assurance provisions shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection of bridge rectifiers. Qualification inspection shall consist of the examinations and tests specified in groups A, B, and C inspections. Tightened inspection for qualification is not required.

4.2.1 Group A inspection. Group A inspection shall consist of the examinations and tests shown in table I, and shall be performed on an inspection sublot of each type.

4.2.2 Group B inspection. Group B inspection shall consist of the examinations and tests shown in table II. Subgroups 1 and 2 of group B inspection may be performed on an inspection sublot of any type to qualify all types. Subgroups 3 and 4 of group B inspection shall be performed on an inspection sublot of the highest-voltage type being qualified.

4.2.3 Group C inspection. Group C inspection shall consist of the examinations and tests specified in table III. Subgroup 1 of group C inspection shall be performed on an inspection sublot of the highest voltage type to qualify that type and all lower-voltage types. Subgroups 2, 3, 4, and 5 of group C inspection may be performed on an inspection sublot of any type to qualify all types. Subgroup 6 of group C inspection shall be performed on an inspection sublot of the highest-voltage type being qualified.

4.3 Quality conformance inspection of bridge rectifiers. Quality conformance inspection shall consist of examinations and tests specified in groups A, B, and C inspections.

4.3.1 Group A inspection. Group A inspection shall consist of the examinations and tests specified in table I. Group A inspection shall be conducted on an inspection sublot of each type.

4.3.2 Group B inspection. Group B inspection shall consist of the examinations and tests specified in table II. Group B inspection may be done on a lot basis with the exception of subgroups 3 and 4. For subgroups 3 and 4, sublots of the highest- and lowest-voltage types present in the lot shall be used to accept those types and all intermediate-voltage types.

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4.3.3 Group C inspection. Group C inspection shall consist of the examinations and tests specified in table III and shall be performed on every tenth lot or every 6 months, whichever comes first. Group C inspection may be performed on a lot basis with the exception of subgroup 1 which shall be performed on a sublot of the highest-voltage type to accept that type and all lower-voltage types. If a lot contains a higher-voltage type than has previously been accepted to subgroup 1 in the current inspection period, this higher-voltage type shall be subjected to subgroup 1 inspection.

4.4 Methods of examination and test. Methods of examination and test shall be as specified in tables I, II, and III, and as follows:

4.4.1 Bridge rectifier assembly.

4.4.1.1 Operational thermal cycling. This test shall be performed by connecting a three phase 60 Hz sine wave input of 10 to 20 volts rms to the ac terminals of the bridge. The resistive load shall be made variable in order to maintain a rated bridge output current of 25 amperes. Bridge case temperature shall be maintained at 55°C except during each cooling cycle when case temperature shall be permitted to drop 20°C maximum. Life-test duration shall be 2,000 cycles with a heating time of $8\ ^{+10}_{-0}$ minutes and a cooling time ≥ 2 minutes.

4.4.1.2 Junction temperature test. To verify that $T_J \leq 175°C$ at maximum rated current and specified case (or ambient) temperature this test shall be performed in accordance with figure 2. Voltage drop V_{F2} shall be read on the trailing edge of the power source voltage waveform (see figure 2b).

4.4.1.3 Dielectric strength test. This test shall be performed with the metal case of the assembly connected to ground and all terminals connected to the high potential side of a dc power supply. The voltage applied between the terminals and the case shall be raised to 2,800 Vdc and shall be held at that level for 10 seconds. Any discontinuity or dynamic instability of the trace, or a current in excess of 10 μAdc shall be cause for rejection.

4.4.1.4 Resistance to solvents. The bridges shall be subjected to tests in accordance with method 215 of MIL-STD-202 and as follows:

(a) All areas of the bridge where marking has been applied shall be brushed.
(b) After completion of the tests, there shall be no evidence of mechanical damage to the device, and markings shall have remained legible.

4.4.1.5 Reverse-recovery time and trace. The reverse-recovery time and trace shall be measured in the circuit of figure 3, or equivalent. The pulse generator shall have a pulse repetition frequency of 1 kHz (maximum) and a pulse width of 10 μs (minimum). The recovery conditions shall be as follows: 0.5 ampere, forward current to 1.0 ampere, reverse current. The recovery trace shall be observed from I = 0 amp to when the diode recovers to -0.25 amp. Recovery trace shall be soft as illustrated on figure 3.

4.4.2 Discrete diodes.

4.4.2.1 Process- and power-conditioning discrete diodes. Discrete diodes for assembly into bridges shall be subjected to the 100-percent inspection specified in table IV, in the order shown. (Discrete diodes shall be devices manufactured on the same production line through final seal, by the same fabrication techniques, and to the same device design with the same material requirements.)

4.4.2.1.1 Screening. The parameters I_R and V_F of table IV shall be measured and the measurements recorded for all diodes in the lot. All diodes shall be identified such that the delta (Δ) end points can be determined after burn-in. All diodes which fail the seal test or fail to meet the limits of I_R and V_F of table IV before burn-in shall be removed from the lot and the quantity removed and type of failure shall be noted on the lot history. All diodes that exceed the delta (Δ) limits of table IV after burn-in shall be removed from the lot and the quantity removed shall be noted on the lot history. If 10 percent or more of the number of diodes subjected to burn-in fail, the entire lot shall be unacceptable.

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TABLE I. Group A inspection

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
Subgroup 1			10				
Visual and mechanical examination	2071			---	---	---	---
Subgroup 2			5				
Forward voltage/leg	4011	i_F = 39 A (pk); t_p = 8.3 ms; duty cycle $\leq$ 2%		V_{FM}	---	1.3	V(pk)
Reverse current/leg	4016	DC method; V_R = rated V_{RWM}		I_R	---	2.0	µAdc
Breakdown voltage/leg	4021	I_R = 50 µAdc (see 4.4.3.1)		BV			
M19500/483-01					240	---	Vdc
02					480	---	Vdc
03					660	---	Vdc
04					880	---	Vdc
Subgroup 3			10				
High-temperature operation:		T_C = 100°C					
Reverse current/leg	4016	DC method; V_R = rated V_{RWM}		I_R	---	200	µAdc

TABLE II. Group B inspection

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
Subgroup 1			15				
Physical dimensions	2066	See figure 1		---	---	---	---
Subgroup 2			10				
Solderability	2026			---	---	---	---
Subgroup 3			10				
Thermal shock (temperature cycling)	1051	Test condition F; 1; time at temperature extremes = 15 minutes (min)		---	---	---	---
Terminal strength:							
Torque	2036	Test cond. D1; 2-inch lbs; t = 10 sec		---	---	---	---
Tension	2036	Test cond. A; t = 30 sec; 10 lbs		---	---	---	---
Moisture resistance	1021	Omit initial conditioning		---	---	---	---

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TABLE II. Group B inspection - Continued

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
Subgroup 3 - Continued							
End points:							
Forward voltage/leg	4011	i_F = 39 A (pk) t_p = ≤ 8.3 ms; duty cycle ≤ 2%		V_{FM}	---	1.3	V(pk)
Reverse current/leg	4016	DC method; V_R = rated V_{RWM}		I_R	---	2.0	µAdc
Subgroup 4			10				
Steady-state operation life	1027	T_A = 150°C; I_O = 0; V_{RWM} = rated		---	---	---	---
End points:							
(Same as subgroup 3)							
Subgroup 5			10				
Operational thermal cycling	---	See 4.4.1.1		---	---	---	---
End points:							
(Same as subgroup 3 plus junction temperature)		See 4.4.1.2 T_C = 55°C; I_O = 25 Adc		T_J	---	175°C	---
Subgroup 6			5				
Dielectric strength test	---	See 4.4.1.3		---	---	---	---

TABLE III. Group C inspection

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
Subgroup 1			20				
Barometric pressure (reduced)	1001	Pressure = 8 mmHg; t = 60 sec		---	---	---	---
Measurement during test:							
Leakage current terminals to case	---	V_{RWM} = rated; All terminals common		---	---	2.0	µAdc
Subgroup 2			20				
Salt atmosphere (corrosion)	1041			---	---	---	---
Subgroup 3			10				
Shock	2016	Nonoperating; 1,500 G; t = 0.5 ms, 5 blows in each orientation X_1, Y_1, and Y_2 (total = 15 blows)		---	---	---	---

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TABLE III. Group C inspection - Continued

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
Subgroup 3 - Continued							
Vibration, variable frequency	2056	Nonoperating; 25 G		---	---	---	---
End points:							
(Same as subgroup 3 of group B)							
Subgroup 4			10				
Resistance to solvents	---	Method 215 of MIL-STD-202 (see 4.4.1.4)		---	---	---	---
Subgroup 5							
Low-temperature operation:		T_A = -65°C					
Forward voltage/leg	4011	i_F = 39 A (pk); duty cycle $\leq$ 2% (pulsed); t_p = 8.3 ms		V_{FM}	---	1.5	V(pk)
Reverse current/leg	4016	DC method; V_R = rated V_{RWM}		I_R	---	2.0	µAdc
Subgroup 6			10				
Surge current	4066	V_{RWM} = rated bridge I_O = 25 Adc; T_C = 55° $^{+10°C}_{-0°C}$; i_f(surge) = 150 A (pk) t_p = 8.3 ms; 10 surges per leg at maximum 1 minute intervals		---	---	---	---
Reverse recovery time and trace (per leg)	---	See 4.4.1.5		---	---	2	µs

4.4.2.2 Quality assurance (lot verification). Quality assurance shall be in accordance with MIL-S-19500, except lot records shall be kept for 1 year, minimum.

4.4.2.3 Burn-in. A half-sine wave of the specified peak voltage shall be impressed across the diode in the reverse direction, followed by a waveform of the specified average rectified current. The forward conduction angle of the rectified current shall be not greater than 180 degrees nor less than 150 degrees.

4.4.2.4 Mounting conditions. The diodes shall be suspended by the leads with at least 6 inches of No. 22 or smaller wire between each lead and the power source. The connecting wires may be soldered to the leads or MS27491-20 contacts may be used. Other clips may be used provided they have equal or less mass than the MS27491-20 contact. No forced air shall be permitted, and the diodes shall be shielded from drafts.

4.4.2.4.1 Alternate mounting conditions. At the option of the manufacturer, other clip or heat sink mounting configurations may be utilized provided that I_O is adjusted such that the junction temperature of each diode is maintained at least 120°C above the specified ambient temperature (T_A).

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4.4.3 Bridge rectifier assembly and discrete diodes.

4.4.3.1 Scope-display evaluation. The reverse breakdown characteristics shall be viewed on an oscilloscope with display calibration factors of 5 to 20 μA/division and 50 to 200 V/division. Reverse current over the knee shall be at least 50 μA. Each diode shall exhibit a sharp knee characteristic and any discontinuity or dynamic instability of the trace shall be cause for rejection.

TABLE IV. Process and power conditioning discrete diodes

Test	MIL-STD-750		Symbol	Limits		Unit
	Method	Details		Min	Max	
High-temperature storage (nonoperating)	---	T_A = 200°C; time = 48 hours (min)	---	---	---	---
Thermal shock (temperature cycling)	1051	Test condition F1; time at temperature extremes = 1 minute (min)	---	---	---	---
Hermetic seal	1071	Test condition G or H for fine leaks; test condition C for gross leaks	---	---	---	---
Forward voltage	4011	i_F = 39 A (pk); t_p = 8.3 ms; duty cycle ≤ 2%	V_{FM}	---	1.3	V(pk)
Reverse current	4016	DC method; V_R = rated V_{RWM}	I_R	---	1.0	μAdc
Scope display evaluation (see 4.4.3.1)	---		---	---	---	---
Burn-in	---	V_{RWM} = rated; T_A = 25°C; I_O = 3 Adc; t = 164 hrs; f = 60 Hz (see 4.4.2.3, 4.4.2.4 and 4.4.2.4.1)	---	---	---	---
End points (within 24 hours of burn-in)						
Forward voltage	4011	i_F = 39 A (pk); t_p = 8.3 ms; duty cycle ≤ 2%	ΔV_F	---	±.05	Vdc
Reverse current	4016	DC method; V_R = rated V_{RWM}	ΔI_R	---	±250	nAdc

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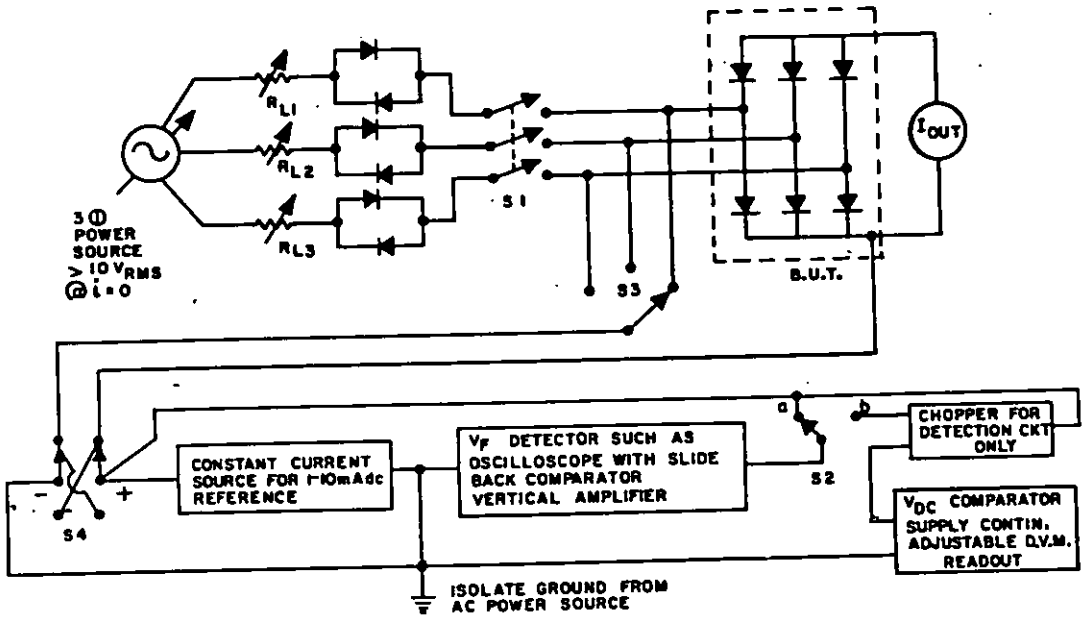


FIGURE 2a. Test circuit.

Procedure:

a. With S1 open, and B.U.T. at 175°C, read V_{F1} at reference current of 10 mAdc.

b. With device held at a case temperature (T_C) of 55°C close S1 and adjust power source and R_L 1, 2, and 3 of Iout 25 Adc. Balance the phase currents by adjusting R_L 1, 2, and 3 to obtain minimum ripple in I_{out}. Establish thermal equilibrium.

c. Observe V_{F2} per figure 2c using the same reference current (±1%) as in a.

d. With S2 at position "a" locate the correct position for reading V_{F2} (see figure2b)

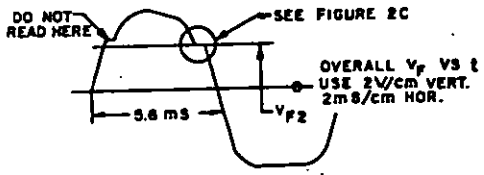


FIGURE 2b. Oscilloscope display.

e. With S2 at position "b" adjust comparator supply to line up V_{F2} with comparator trace (see figure 2c)! Read V_{F2} on digital voltmeter.

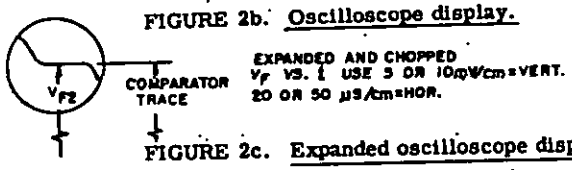


FIGURE 2c. Expanded oscilloscope display.

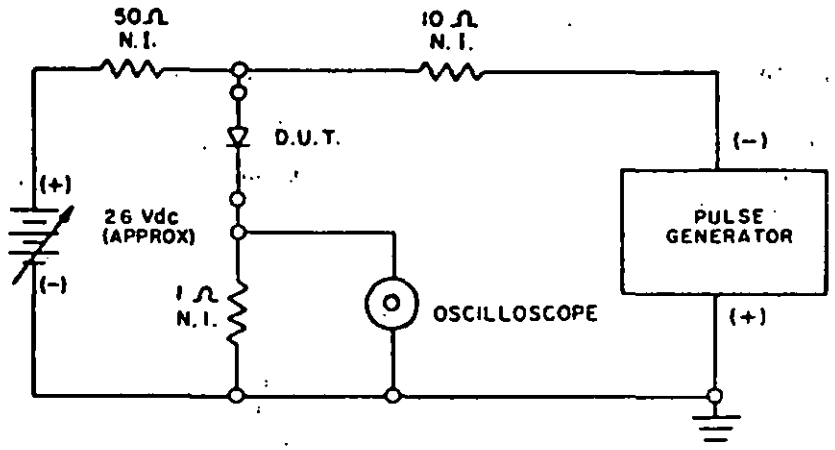
f. Adjust S3 and S4 so that all diodes in the B.U.T. shall be tested.

Acceptance criterion:

a. V_{F2} shall be $\geq V_{F1}$. (Indicated $T_J \leq 175°C$.)

FIGURE 2. Junction temperature test for bridge rectifier.

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NOTES:
1. Oscilloscope: Rise time ≤ 7 ns input impedance = 1 Meg; 22 pF.
2. Pulse Generator: Rise time ≤ 10 ns source impedance = 50Ω.

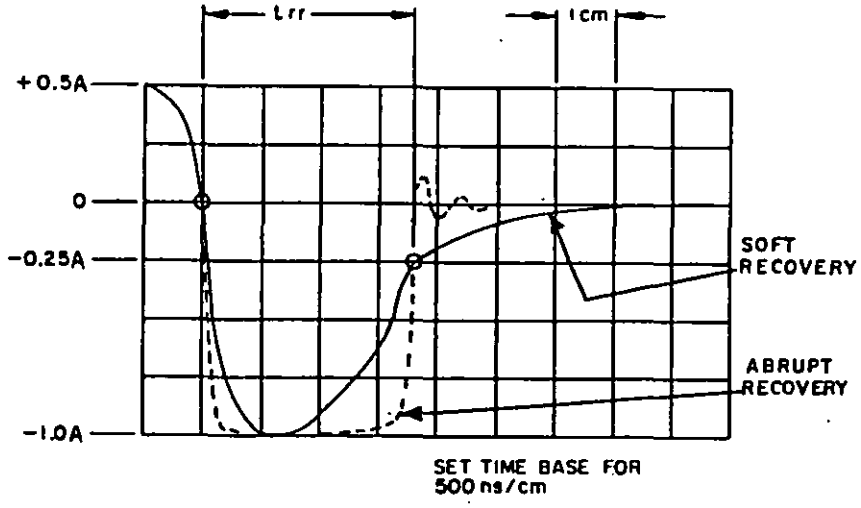


FIGURE 3. Reverse recovery test circuit and characteristic waveform.

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5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with MIL-S-19500.

6. NOTES

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Application data. Design engineers who procure three phase full wave bridges to this specification shall provide adequate heat dissipation to insure that case temperature is maintained at 55°C when operating the bridge at full rated current of 25 amperes. At this current, the bridge total power dissipation will be approximately 50 watts.

Custodians:
Army - EL
Navy - EC
Air Force - 17

Review activities:
Army - MI, MU
Air Force - 11, 80
DSA - ES
NASA - NA

User activities:
Army - SM
Navy - CG, MC, OS, AS, SH
Air Force - 19

Preparing activity:
Navy - EC

Agent:
DSA - ES

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   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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